Cyclohexane Reactor Ver 1 · operator training simulator

Pre-commissioning and commissioning plan

Document no. CHX-V1-CM-PLN-001 · **Discipline** Commissioning

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- 7 systems / 24 subsystems; 29 mechanical-completion and pre-commissioning items, 6 test packs, 12 loop checks, 8 motor solo runs.
- 17 SIS function tests and 28 alarm tests: one for every row of the C&E; CHX-V1-IN-CE-001.
- PSSR: 13 items; 77 of 82 HAZOP and Safety-layers recommendations to close before start-up.
- 21 commissioning and start-up steps mapped to the SOPs and the graded procedures of the OTS, timed by the reference runs; performance test with 11 acceptance criteria from the model at design; handover.
- Companion document: procurement packages CHX-V1-PC-PKG-001.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Purpose, scope and definitions

This plan takes the cyclohexane unit from the end of construction to handover. It is written from the simulator model and the other teaching documents and is checked against them in the "Procurement to commissioning" tab of the OTS. Durations not taken from the model are labelled typical or assumed.

Term	Meaning in this plan
Mechanical completion (MC)	A subsystem installed as the P&ID, pressure-tested, with no punch-A item open; certified per subsystem.
Pre-commissioning	Making a subsystem clean, tight, dry and proven without process fluids: flushing, blowing, pressure and leak tests, drying, loop checks, SIS function tests, motor runs, catalyst loading.
Commissioning	Bringing utilities and inert gas into the systems and proving them working: nitrogen inerting and tightness, utilities in service.
PSSR / RFSU	Pre-start-up safety review (OSHA 29 CFR 1910.119(i)); its signature is the ready-for-start-up milestone: hydrogen and hydrocarbon may be admitted.
Start-up	First hydrocarbon and hydrogen to on-spec product at design, following SOP-CYC-01, -02, -07 and -10.
Performance test	Test run at design against the acceptance criteria of section 8; passed → provisional acceptance (typical).
Punch A / B	A: must be closed before start-up; B: may be closed afterwards with an owner and a date.

Phases and milestones

#	Phase	What is done	Output / milestone
1	Procure	Packages bought, inspected and expedited; long-lead items ordered first	POs, inspection release notes, vendor data books
2	Construct	Civil, erection, piping, electrical and instrument installation	Installation records, test packs
3	Mechanical completion	System by system: installed as the P&ID, tested, punch-A items closed	MC certificate per system
4	Pre-commissioning	Flushing, pressure and leak tests, drying, loop checks, SIS function tests, motor runs, catalyst loading	Test records; ready for commissioning
5	Commissioning	Utilities in, loop inerted and tight; PSSR before hydrocarbon	PSSR signed: ready for start-up (RFSU)
6	Start-up	First hydrocarbon and hydrogen, light-off, ramp to design, on-spec product	Start-up complete (line-out)
7	Performance test	Test run at design against the acceptance criteria	Performance test report; provisional acceptance
8	Handover	Care, custody and control to operations with the documents and the punch-B list	Handover certificate

2 · System breakdown

Commissioning is done by system. Every test record, punch item and certificate carries its subsystem number.

System	Subsystem	Name	Tags	Description
S10 Utilities	S10.1	Cooling water	HV-113, E-103 (CW side), E-106 (CW side), R-102 jacket (COOL), TI-120	Supply 32 °C to E-103, E-106 and the R-102 jacket in COOL.
S10 Utilities	S10.2	MP steam 10 bar g	HV-110, HS-101, HS-102, TV-102	Start-up heating of R-101 through the E-101 shell (HS-101 HEAT) and the R-102 jacket (HS-102 HEAT).
S10 Utilities	S10.3	Boiler feed water and LP steam	D-101, LIC-109, LV-109, PIC-110, PV-110, FI-109, FI-111	D-101 raises LP steam from the R-101 reaction heat and exports it to the 3.5 bar g header.
S10 Utilities	S10.4	Nitrogen	HV-120	Purging, inerting, leak testing, catalyst blanketing.
S10 Utilities	S10.5	Instrument air	IA header	Air to every control valve and SIS valve; loss of air sends valves to their fail positions.
S10 Utilities	S10.6	Electrical power	MCC, P-100A/B, P-101A/B, P-102A/B, M-101, K-101	Power to the eight motors of the model.
S10 Utilities	S10.7	Flare and vent	HV-121, EDV-101, FL-100	Depressuring and venting to flare; flare purge and knock-out (HAZOP R30, SL-R23).
S20 Benzene feed	S20.1	Benzene feed to R-101	P-100A, P-100B, FIC-101, FV-101, XV-101	From TK-100 at the battery limit to the R-101 inlet.
S30 Hydrogen loop	S30.1	Make-up hydrogen	PIC-105, PV-105, XV-105, FI-106	99.5 % hydrogen into V-104 on loop-pressure control.
S30 Hydrogen loop	S30.2	Recycle compressor	K-101, FIC-103, FV-103, PI-112	V-104 gas back to the R-101 sparger.
S30 Hydrogen loop	S30.3	Purge and purity	HIC-107, FI-107, AI-105	Purge to fuel gas holds the recycle purity.
S30 Hydrogen loop	S30.4	Loop oxygen analyser	AI-100	Oxygen in the loop before hydrogen is admitted.
S40 R-101 reaction	S40.1	R-101 and agitator	R-101, M-101, TI-100, TIC-101, PI-101, LIC-101, LV-101, JI-101, HV-116, HV-117, catalyst pot	Slurry CSTR with Raney-Ni at 200 °C and 30 bar g.
S40 R-101 reaction	S40.2	Pump-around and E-101	P-101A, P-101B, E-101, FIC-104, FV-104, TI-104	Reaction heat to D-101 through E-101.
S40 R-101 reaction	S40.3	Overhead condenser and reflux	E-103, V-103, FI-105, TI-103	Boil-up condensed on cooling water and refluxed to R-101.
S50 R-102 and catalyst filtration	S50.1	R-102	R-102, TIC-102, TV-102, LIC-102, LV-102	Second slurry CSTR at 160 °C with a jacket.
S50 R-102 and catalyst filtration	S50.2	Transfer pumps and filters	P-102A, P-102B, F-102A, F-102B, PDI-102, FI-102	Slurry pumped through the candle filter; catalyst stays behind.
S60 Finishing and separation	S60.1	R-103 finishing reactor	R-103, TI-106, TI-107A…E	Trickle bed of supported Ni.
S60 Finishing and separation	S60.2	Product cooler	E-106, TI-109	R-103 effluent cooled on cooling water.
S60 Finishing and separation	S60.3	HP separator	V-104, LIC-104, LV-104, LT-104S, PI-105	Gas to K-101, liquid product to storage.
S60 Finishing and separation	S60.4	Product routing and analysers	HS-115, FI-110, AI-103, AI-104, TK-106, TK-107	On-spec product to TK-106, off-spec to TK-107.
S70 Control and safety systems	S70.1	DCS	DCS, all control loops	Basic process control and alarms.
S70 Control and safety systems	S70.2	SIS	SIS-100, HS-100, first-out panel	Trips, permissives and the ESD.
S70 Control and safety systems	S70.3	Fire and gas, emergency response	GD-1…3, FD-1, deluge, fire water	Recommended additions SL-R29 … SL-R44.

3 · Mechanical completion and pre-commissioning checklist

Mechanical completion (5)

ID	Subsystem	Item	Acceptance	Done / sign
MC-01	S70.1	Walk-down of every system against the P&ID CHX-V1-PID-001; punch list raised (A = before start-up, B = may follow)	No punch-A item open on the system	
MC-02	S10.6	Earthing, cable insulation resistance and continuity, hazardous-area (Ex) inspection against CHX-V1-EE-HAC-001	Records signed for every circuit	
MC-03	S30.2	K-101 and M-101 alignment checked by the vendor representative after grouting and piping hook-up (piping strain)	Alignment within the vendor tolerance; record	
MC-04	S10.7	Relief valves installed with set certificates and seals (recommended additions SL-R8 ... SL-R16; see the Relief & vent tab)	Tag, set pressure and certificate agree with the relief load basis	
MC-05	S70.2	Instruments installed and calibrated; SIS sensors on their own impulse lines / elements as the SRS requires	Calibration certificate per instrument	

Flushing and blowing (9)

ID	Subsystem	Item	Acceptance	Done / sign
FL-01	S20.1	Benzene feed lines water-flushed with temporary strainers at P-100A/B suction	Clean water at the outlet; strainers clean after the last run	
FL-02	S40.2	Pump-around loop water-flushed with E-101 tubes bypassed by a jumper (no debris into the exchanger)	Clean outlet; jumper removed, E-101 reconnected	
FL-03	S50.2	R-102 → P-102 → F-102 → R-103 flushed with the filter candles out and R-103 bypassed (before catalyst)	Clean outlet; candles installed after the flush	
FL-04	S30.2	Gas loop blown with air or nitrogen (V-104 → K-101 → FV-103 → sparger; R-101 overhead → E-103 → V-103 → R-102 → R-103 → E-106 → V-104), K-101 by a spool, sparger blinded	Target plates clean (typical: no impacts on two consecutive plates)	
FL-05	S30.2	K-101 lube-oil and seal systems oil-flushed	Cleanliness to the compressor package requirement	
FL-06	S10.2	MP steam to E-101 (HV-110) and to the R-102 jacket blown with steam	Target plates clean on consecutive blows (typical)	
FL-07	S10.3	D-101 and E-101 shell side alkaline boil-out / chemical clean before raising steam (typical for steam generators)	Water quality to the boiler-water specification	
FL-08	S10.1	Cooling-water lines flushed with E-103, E-106 and the R-102 jacket isolated until the water runs clean	Clean outlet	
FL-09	S10.4	Nitrogen and instrument-air lines blown dry and clean	Dew point and cleanliness at the end user	

Pressure tests (6)

ID	Subsystem	Item	Acceptance	Done / sign
TP-01	S40.1	Hydrogen loop with reactors and vessels (R-101, E-103/V-103, R-102, R-103 before catalyst, E-106 process side, V-104, F-102A/B): water (hydrotest) at 46.8 bar g (design 36 bar g × 1.3; vessels in the pack limit the test to 1.3 × design (ASME VIII-1 UG-99, typical))	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	
TP-02	S30.2	K-101 discharge piping to FV-103 (compressor casing blinded off): nitrogen (pneumatic) at 46.2 bar g (design 42 bar g × 1.1; pneumatic 1.1 × design (ASME B31.3 345.5) keeps water out of the compressor piping; exclusion zone and approval (typical))	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	
TP-03	S40.2	Pump-around loop P-101A/B → FV-104 → E-101 tube side → R-101: water (hydrotest) at 46.8 bar g (design 36 bar g × 1.3; E-101 in the pack: 1.3 × design (typical))	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	

ID	Subsystem	Item	Acceptance	Done / sign
TP-04	S20.1	Benzene feed P-100A/B → FV-101 → XV-101 → R-101 inlet: water (hydrotest) at 54 bar g (design 36 bar g × 1.5; piping only: 1.5 × design (ASME B31.3 345.4); design pressure assumed equal to the loop, confirm on the line list)	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	
TP-05	S10.3	Steam side: D-101, E-101 shell, LP steam export to PV-110: water (hydrotest) at 10.4 bar g (design 8 bar g × 1.3; 1.3 × design; design pressure taken as the assumed PSV-109 set — see EQ-ec-2 (MP steam at 10 bar g reaches this shell))	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	
TP-06	S10.1	Cooling-water sides of E-103 and E-106 and the R-102 jacket: water (hydrotest) at 13 bar g (design 10 bar g × 1.3; design pressure taken as the assumed thermal-relief set; 1.3 × design)	Hold 30 min (typical) with no leak and no pressure loss other than temperature; record signed	

Worked example. TP-01: design 36 bar g (Safety-layers assumption = PSV set) × 1.3 = **46.8 bar g** hydrotest. TP-02: design 42 bar g × 1.1 = **46.2 bar g** pneumatic — a gas test stores far more energy than a water test at the same pressure, so it needs an exclusion zone and approval.

Leak tests (3)

ID	Subsystem	Item	Acceptance	Done / sign
LT-01	S30.1	Loop nitrogen tightness test at the purge pressure (≈ 7.6 bar g, as in the reference purge of the Transition tab) (SOP-CYC-01)	Pressure loss below 0.1 bar in 30 min (typical; SOP-CYC-01)	
LT-02	S30.1	Nitrogen with a helium tracer (typical 1 %) at the operating pressure 29 bar g: every flange, valve gland, the agitator seal and the K-101 seals sniffed	No detectable tracer at any joint (typical)	
LT-03	S40.1	M-101 double-seal and K-101 seal-gas leak rates at nitrogen pressure	Within the vendor limits	

Drying (3)

ID	Subsystem	Item	Acceptance	Done / sign
DR-01	S40.1	Hydrotested loop drained at the low points and dried with dry nitrogen (free water would flash at 200 °C and end in V-104 and the product)	Dew point at the vents ≤ −20 °C (assumed, typical)	
DR-02	S60.1	R-103 dried before the supported-Ni catalyst is loaded	Dew point ≤ −20 °C (assumed)	
DR-03	S30.2	K-101 casing and suction/discharge piping dry before the first run	No free water at the drains	

Catalyst loading (3)

ID	Subsystem	Item	Acceptance	Done / sign
CL-01	S60.1	R-103: inert balls, supported-Ni catalyst and top inerts loaded (sock or dense loading, typical) with mass and outage recorded per layer; bed ΔP checked with nitrogen; R-103 closed and put under nitrogen	Loaded mass and outage as the loading diagram	
CL-02	S40.1	Raney-Ni drums received under liquid, inspected (liquid above the catalyst, no bulging) and stored wet in the shade (SOP-CYC-07)	Every drum inspected; SDS at the store	
CL-03	S40.1	Raney-Ni transferred to the catalyst pot under a nitrogen blanket, never dry; the carrier liquid exchanged to cyclohexane before the charge (EQ-ec-5); pot closed and purged (SOP-CYC-07)	Pot under nitrogen, O_2 below 1 % (SOP-CYC-07 hold point); charging to R-101 happens at start-up step C-05	

Instrument loop checks (12)

Loop	Service	Range	Fail	Subsystem	Check	Done / sign
FIC-101	Benzene feed to R-101 → FV-101	0–60 t/h	fail closed	S20.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action reverse.	

Loop	Service	Range	Fail	Subsystem	Check	Done / sign
FIC-103	Hydrogen (recycle + make-up) to R-101 sparger → FV-103	0–80000 Nm³/h	fail closed	S30.2	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action reverse.	
PIC-105	Loop pressure at V-104 → make-up hydrogen PV-105	0–40 bar g	fail closed	S30.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action reverse.	
TIC-101	R-101 temperature → cascade to FIC-104 pump-around through E-101	0–300 °C	—	S40.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; output to the slave loop; action direct.	
FIC-104	Pump-around flow through E-101 → FV-104	0–700 t/h	fail open	S40.2	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail open on air loss; action reverse.	
LIC-101	R-101 level → LV-101 overflow to R-102	0–100 %	fail closed	S40.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action direct.	
TIC-102	R-102 temperature → TV-102 jacket (MP steam in HEAT, CW in COOL)	0–300 °C	fail closed	S50.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action reverse.	
LIC-102	R-102 level → LV-102 to R-103 through filter F-102	0–100 %	fail closed	S50.1	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action direct.	
LIC-104	V-104 separator level → LV-104 product to storage	0–100 %	fail closed	S60.3	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action direct.	
LIC-109	D-101 steam drum level → LV-109 boiler feed water	0–100 %	fail open	S10.3	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail open on air loss; action reverse.	
PIC-110	D-101 steam pressure → PV-110 LP steam export	0–10 bar g	fail open	S10.3	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail open on air loss; action direct.	
HIC-107	Purge from V-104 to fuel gas (hand controller)	0–100 %	fail closed	S30.3	Transmitter 0-25-50-75-100 % from the field to the DCS faceplate; valve stroked from the DCS (MAN 0/50/100 %) and seen moving in the field; fail closed on air loss; action reverse.	

Motor solo runs (8)

Motor	Subsystem	Solo and coupled run	Start interlock in the OTS	Done / sign
P-100A	S20.1	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	—	
P-100B	S20.1	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	—	
P-101A	S40.2	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	start blocked below 20 % R-101 level	
P-101B	S40.2	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	start blocked below 20 % R-101 level	
P-102A	S50.2	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	start blocked below 8 % R-102 level	
P-102B	S50.2	Uncoupled motor run 2 h (typical): rotation, vibration, bearing temperatures, current; coupled run on flushing liquid with the suction strainer in	start blocked below 8 % R-102 level	
K-101	S30.2	Uncoupled motor run 2 h (typical); coupled run under the vendor representative on nitrogen at the reduced speed or suction pressure the vendor allows (nitrogen is about 14 times denser than hydrogen: motor overload)	start blocked below 10 bar g suction, above 85 % V-104 level or with the SIS tripped	

Motor	Subsystem	Solo and coupled run	Start interlock in the OTS	Done / sign
M-101	S40.1	Uncoupled motor run 2 h (typical); coupled run only with R-101 filled with start-up cyclohexane above the interlock level (never dry: seal and impeller)	start blocked below 25 % R-101 level	

4 · SIS function tests and alarm tests against the C&E

Every row of the C&E; CHX-V1-IN-CE-001 gets one test. Trip rows: the cause is forced at the instrument and held for the delay; every marked effect is checked in the field; the trip latches, the first-out shows the cause and the reset is refused while the cause is present. Permissive rows: with the cause present the open or start command must be refused. Where the SRS CHX-V1-IN-SRS-001 gives a proof-test method it is used as the first test. XS-101 and XS-102 are armed only with XV-101 open, which needs every start permissive: before hot commissioning they are tested by simulating the inputs at the logic solver under a test permit, and proven again at the first start.

SIS function tests (17)

Test	C&E cause	Setting	Delay	Effects to prove	Method	Acceptance	Sign
SFT-01 SIF-7	HS-100 — Emergency shutdown pushbutton (operator)	manual	0 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, EDV-101, 1ST-OUT	SRS proof test used as the first test: press HS-100 on a shutdown; XV-101, XV-105 close, EDV-101 opens (stroke time), K-101 stops.	After 0 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · EDV-101 O · 1ST-OUT FO; latched, first-out shows HS-100; reset refused while the cause is present.	
SFT-02 SIF-1	TSHH-100 — R-101 temperature high-high	220 °C	5 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, 1ST-OUT	SRS proof test used as the first test: inject at the element (calibrator bath or simulated signal at the transmitter), trip at 220 °C after 5 s; confirm XV-101 and XV-105 close within 2 s and K-101 stops; check the first-out.	After 5 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · 1ST-OUT FO; latched, first-out shows TSHH-100; reset refused while the cause is present.	
SFT-03 SIF-2	TSHH-107 — R-103 bed temperature high-high	220 °C	5 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, 1ST-OUT	SRS proof test used as the first test: each of the five elements in turn, trip at 220 °C after 5 s; valve stroke and K-101 stop as for SIF-1. Compare the five cells on every shift (at design they agree within a fraction of a degree) to find a drifting element.	After 5 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · 1ST-OUT FO; latched, first-out shows TSHH-107; reset refused while the cause is present.	
SFT-04 SIF-3	PSHH-101 — R-101 pressure high-high	33.5 bar g	3 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, 1ST-OUT	SRS proof test used as the first test: pressure applied at PT-101, trip at 33.5 bar g after 3 s; XV-105 stroke-time and seat-leak test (it is the only make-up isolation); K-101 stop.	After 3 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · 1ST-OUT FO; latched, first-out shows PSHH-101; reset refused while the cause is present.	
SFT-05 SIF-4	XS-101 — Agitator M-101 stopped with XV-101 open	stopped	10 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, 1ST-OUT	SRS proof test used as the first test: stop M-101 with XV-101 open on a test permit (or simulate the contact), trip after 10 s; valve stroke and K-101 stop.	After 10 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · 1ST-OUT FO; latched, first-out shows XS-101; reset refused while the cause is present.	
SFT-06 SIF-5	XS-102 — Compressor K-101 stopped with XV-101 open	stopped	10 s	XV-101, XV-105, K-101, FIC-101, FIC-103, PIC-105, HS-101, FV-104, HS-102, 1ST-OUT	SRS proof test used as the first test: stop K-101 with XV-101 open on a test permit (or simulate the contact), trip after 10 s; XV-101 stroke time.	After 10 s: XV-101 C · XV-105 C · K-101 STOP · FIC-101 MAN 0 % · FIC-103 MAN 0 % · PIC-105 MAN 0 % · HS-101 COOL · FV-104 FO 100 % · HS-102 COOL · 1ST-OUT FO; latched, first-out shows XS-102; reset refused while the cause is present.	

Test	C&E cause	Setting	Delay	Effects to prove	Method	Acceptance	Sign
SFT-07 SIF-6	LSHH-104 — V-104 level high-high: liquid toward K-101	90 %	5 s	K-101	SRS proof test used as the first test: raise the level on LT-104S (or simulate it), K-101 stops after 5 s; check the restart block above 85 %.	After 5 s: K-101 STOP; not latched, no first-out.	
SFT-08	SIS-100 — SIS trip latched (not reset)	latched	0 s	XV-101, XV-105, K-101	Hold the cause present (SIS trip latched (not reset); setting latched) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101, XV-105, K-101 while the cause is present; accepted when it clears.	
SFT-09 PERM-H2	AI-100 — Loop oxygen high before hydrogen	1 %	0 s	XV-101, XV-105	SRS proof test used as the first test: analyser calibrated with zero and span gas; with test gas above 1 % the open command for XV-105 must be refused; XV-105 seat-leak test.	Command refused for XV-101, XV-105 while the cause is present; accepted when it clears.	
SFT-10	AI-105 — H2 not admitted or loop purity low	80 %	0 s	XV-101	Hold the cause present (H2 not admitted or loop purity low; setting 80 %) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-11	LIC-101 — R-101 level low (agitator, sparger covered)	40 %	0 s	XV-101	Hold the cause present (R-101 level low (agitator, sparger covered); setting 40 %) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-12	M-101 — Agitator not running	stopped	0 s	XV-101	Hold the cause present (Agitator not running; setting stopped) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-13	FIC-103 — K-101 off or H2 flow to sparger low	15,000 Nm³/h	0 s	XV-101	Hold the cause present (K-101 off or H2 flow to sparger low; setting 15000 Nm³/h) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-14	TI-100 — R-101 below light-off temperature	140 °C	0 s	XV-101	Hold the cause present (R-101 below light-off temperature; setting 140 °C) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-15	HV-113 — E-103 reflux condenser CW not open	50 % open	0 s	XV-101	Hold the cause present (E-103 reflux condenser CW not open; setting 50 % open) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for XV-101 while the cause is present; accepted when it clears.	
SFT-16	PIC-105 — K-101 suction pressure low	10 bar g	0 s	K-101	Hold the cause present (K-101 suction pressure low; setting 10 bar g) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for K-101 while the cause is present; accepted when it clears.	
SFT-17	LT-104S — V-104 level high, liquid toward K-101	85 %	0 s	K-101	Hold the cause present (V-104 level high, liquid toward K-101; setting 85 %) and give the open / start command: it must be refused; remove the cause and repeat: it must be accepted.	Command refused for K-101 while the cause is present; accepted when it clears.	

Alarm tests (28)

Test	C&E alarm	Description	Setting	DCS tag	Subsystem	Sign
ALT-01	FAL-101	Benzene feed low	25 t/h	FIC-101	S20.1	
ALT-02	FAH-101	Benzene feed high	42 t/h	FIC-101	S20.1	
ALT-03	FAL-103	Hydrogen to R-101 sparger low	35,000 Nm³/h	FIC-103	S30.2	
ALT-04	PAL-105	Loop pressure (V-104) low	27.5 bar g	PIC-105	S30.1	
ALT-05	PAH-105	Loop pressure (V-104) high	30.5 bar g	PIC-105	S30.1	
ALT-06	TAL-101	R-101 temperature low	190 °C	TIC-101	S40.1	
ALT-07	TAH-101	R-101 temperature high	208 °C	TIC-101	S40.1	
ALT-08	TAH-100	R-101 temperature (independent element) high	210 °C	TI-100	S40.1	
ALT-09	LAL-101	R-101 level low	45 %	LIC-101	S40.1	
ALT-10	LAH-101	R-101 level high	82 %	LIC-101	S40.1	
ALT-11	JAL-101	Agitator M-101 power low	80 kW	JI-101	S40.1	
ALT-12	TAL-102	R-102 temperature low	148 °C	TIC-102	S50.1	
ALT-13	TAH-102	R-102 temperature high	172 °C	TIC-102	S50.1	
ALT-14	LAL-102	R-102 level low	30 %	LIC-102	S50.1	
ALT-15	LAH-102	R-102 level high	80 %	LIC-102	S50.1	
ALT-16	PDAH-102	Catalyst filter F-102 ΔP high	0.8 bar	PDI-102	S50.2	
ALT-17	TAH-107	R-103 hottest bed temperature high	185 °C	TI-107	S60.1	
ALT-18	LAL-104	V-104 separator level low	25 %	LIC-104	S60.3	
ALT-19	LAH-104	V-104 separator level high	78 %	LIC-104	S60.3	
ALT-20	AAH-103	Product benzene (spec 10 ppm) high	10 ppm	AI-103	S60.4	
ALT-21	AAH-104	Product MCP isomer (spec 200 ppm) high	200 ppm	AI-104	S60.4	
ALT-22	AAL-105	Recycle hydrogen purity low	85 % H2	AI-105	S30.3	
ALT-23	LAL-109	D-101 steam drum level low	25 %	LIC-109	S10.3	
ALT-24	LAH-109	D-101 steam drum level high	75 %	LIC-109	S10.3	
ALT-25	PAL-110	D-101 steam pressure low	3.8 bar g	PIC-110	S10.3	
ALT-26	PAH-110	D-101 steam pressure high	5.6 bar g	PIC-110	S10.3	
ALT-27	TAH-103	E-103 vent temperature (reflux condenser) high	75 °C	TI-103	S40.3	
ALT-28	AAH-100	Loop oxygen with hydrogen admitted high	1 % O2	AI-100	S30.4	

Alarm test method: inject at the transmitter (or simulate in the DCS) beyond the setting; acceptance: DCS alarm and horn at the setting, clearing with its deadband.

5 · Pre-start-up safety review (PSSR)

Item	Requirement (1910.119)	Check	Where	Done / sign
PS-01	(i)(2)(i) Construction and equipment as designed	Mechanical completion certificate signed for every subsystem of part 5; no punch-A item open	part 6 MC-01 … MC-05	
PS-02	(i)(2)(i) Construction and equipment as designed	Pressure-test (TP-01 … TP-06) and leak-test (LT-01 … LT-03) records reviewed and signed	part 6	
PS-03	(i)(2)(i) Construction and equipment as designed	Relief devices in place with set certificates; relief load basis issued	Relief & vent tab, CHX-V1-PE-RLB-001	
PS-04	(i)(2)(i) Construction and equipment as designed	Every SIS function test signed, all overrides and forces removed, SIS reset and in normal mode	part 7, C&E CHX-V1-IN-CE-001	
PS-05	(i)(2)(i) Construction and equipment as designed	Every alarm test signed; analysers calibrated (AI-100 is a permissive)	part 7, PK-12	
PS-06	(i)(2)(i) Construction and equipment as designed	Utilities in service and stable: cooling water, MP steam, BFW, nitrogen, instrument air, power, flare with its purge	part 6, SOP-CYC-01 prerequisites	
PS-07	(i)(2)(ii) Procedures in place and adequate	Operating, emergency and safe-work procedures approved and at the console	SOP tab register · all 13 SOPs	
PS-08	(i)(2)(ii) Procedures in place and adequate	Emergency response: fire water, monitors, deluge, gas and flame detection, safety showers in service; muster drill held	SL-R29 … SL-R44	
PS-09	(i)(2)(ii) Procedures in place and adequate	Raney-Ni handling: procedure, PPE, water hose and wet-sand kit at the catalyst pot	HAZOP R31 · SOP-CYC-07	
PS-10	(i)(2)(iii) Hazard analysis recommendations resolved	HAZOP and Safety-layers recommendations marked "before start-up" closed; the others have an interim measure and an MOC (table below)	HAZOP tab, Safety layers tab	
PS-11	(i)(2)(iv) Training completed	Every console operator has passed the graded cold startup, normal shutdown, pump-around trip and filter change-over on the simulator	Exercises tab, Debrief tab	
PS-12	(l) Management of change	Field changes since the HAZOP reviewed under MOC; as-built P&ID marked up	Project & deliverables tab	
PS-13	Site readiness	Permits closed, blinds removed or listed on the blind register, scaffolds and temporary equipment removed, area clean	typical practice · SOP-CYC-12	

HAZOP and Safety-layers recommendations to close before start-up

Disposition: **SIS** Changes the trip or alarm logic the start-up relies on: configure and function-test before hydrogen or benzene is admitted. · **HW** Physical protection that must exist when hydrocarbon and hydrogen are first introduced. · **DOC** A study or procedure the start-up steps depend on: approve before start-up (and build whatever it concludes). · **POST** Improves monitoring or reliability; the start-up does not depend on it. Interim measure (operator round, manual action in the SOP) recorded under MOC, close within the first months.
Before start-up: R01, R02, R03, R04, R05, R06, R07, R08, R09, R11, R13, R14, R16, R17, R19, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, SL-R1, SL-R2, SL-R3, SL-R4, SL-R5, SL-R6, SL-R7, SL-R8, SL-R9, SL-R10, SL-R11, SL-R12, SL-R13, SL-R14, SL-R15, SL-R16, SL-R17, SL-R18, SL-R19, SL-R20, SL-R21, SL-R22, SL-R23, SL-R24, SL-R25, SL-R26, SL-R27, SL-R28, SL-R29, SL-R30, SL-R31, SL-R32, SL-R33, SL-R34, SL-R35, SL-R36, SL-R37, SL-R38, SL-R39, SL-R40, SL-R41, SL-R42, SL-R43, SL-R44.
May follow start-up (interim measure under MOC): R10 — Interim: the spare pump is started by the operator (SOP-CYC-09, graded "pump-around trip").; R12 — Interim: daily check of the FIC-104 output and E-101 performance by the process engineer.; R15 — Interim: blowdown sample in the shift routine.; R18 — Interim: product routing by HS-115 after two on-spec analyses (SOP-CYC-02).; R20 — Interim: product fines checked by the laboratory each shift.

ID	Source	Recommendation	Disposition	Before start-up	Package	Closed / sign
R01	HAZOP	Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.	SIS	yes	PK-10	
R02	HAZOP	Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.	DOC	yes	PK-15	

ID	Source	Recommendation	Disposition	Before start-up	Package	Closed / sign
R03	HAZOP	Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).	SIS	yes	PK-10	
R04	HAZOP	Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.	SIS	yes	PK-10	
R05	HAZOP	Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).	SIS	yes	PK-10	
R06	HAZOP	Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.	SIS	yes	PK-10	
R07	HAZOP	Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).	SIS	yes	PK-10	
R08	HAZOP	Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.	HW	yes	PK-06	
R09	HAZOP	Analyse every benzene receipt for sulphur and water; drain TK-100 water; consider a guard bed ahead of R-101.	DOC	yes	—	
R10	HAZOP	Auto-start the standby pumps P-100B, P-101B and P-102B on low discharge pressure or flow.	POST	no	—	
R11	HAZOP	Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).	SIS	yes	PK-11	
R12	HAZOP	Add a control-margin alarm on the FIC-104 output and trend the E-101 heat-transfer coefficient.	POST	no	—	
R13	HAZOP	Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.	DOC	yes	PK-15	
R14	HAZOP	Add low-low and high-high level on D-101 independent of LIC-109.	SIS	yes	PK-10	
R15	HAZOP	Monitor D-101 blowdown or condensate for hydrocarbon (E-101 tube leak).	POST	no	—	
R16	HAZOP	Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.	SIS	yes	PK-10	
R17	HAZOP	Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).	HW	yes	PK-12	
R18	HAZOP	Route product to TK-107 automatically on AI-103 or AI-104 high and alarm a stale analyser value.	POST	no	—	
R19	HAZOP	Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.	SIS	yes	PK-10	
R20	HAZOP	Add a low ΔP alarm on PDI-102 (torn candle) and a routine fines check of the product.	POST	no	—	
R21	HAZOP	Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.	SIS	yes	PK-11	
R22	HAZOP	Add a high alarm on TI-109 (V-104 temperature).	SIS	yes	PK-11	
R23	HAZOP	Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.	SIS	yes	PK-11	
R24	HAZOP	Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.	SIS	yes	PK-10	
R25	HAZOP	Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).	HW	yes	PK-17	
R26	HAZOP	Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.	SIS	yes	PK-10	
R27	HAZOP	Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.	HW	yes	PK-07	
R28	HAZOP	Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.	HW	yes	PK-14	
R29	HAZOP	Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.	DOC	yes	PK-10	
R30	HAZOP	Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.	HW	yes	PK-14	
R31	HAZOP	Write a Raney-Ni handling procedure: always wet and under nitrogen, metered charges through HV-117, filter-cake handling.	DOC	yes	—	

ID	Source	Recommendation	Disposition	Before start-up	Package	Closed / sign
R32	HAZOP	Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.	DOC	yes	PK-10	
R33	HAZOP	Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).	SIS	yes	PK-11	
R34	HAZOP	Gas-free test and positive isolation before opening any equipment (filters, reactors); written filter change-over procedure.	DOC	yes	—	
R35	HAZOP	Fit minimum-flow bypasses on P-100A/B and P-102A/B.	HW	yes	PK-06	
R36	HAZOP	Review the blanketing and venting of TK-106 / TK-107 for hydrogen flashing from the product.	DOC	yes	—	
R37	HAZOP	Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.	SIS	yes	PK-11	
R38	HAZOP	Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.	HW	yes	PK-14	
SL-R1	Safety layers	R-101 pressure pre-alarm on PT-101 (PIAH-101, H 32.5 bar g (assumed))	SIS	yes	PK-11	
SL-R2	Safety layers	Deviation alarm TIC-101 element vs TI-100 element (TDA-101, |ΔT| > 5 °C (assumed))	SIS	yes	PK-11	
SL-R3	Safety layers	D-101 level low-low → unit trip (LSLL-109, 10 % (assumed))	SIS	yes	PK-10	
SL-R4	Safety layers	Agitator shaft-speed switch, voted with the M-101 run signal (SSL-101, low speed (assumed 50 % of design))	SIS	yes	PK-10	
SL-R5	Safety layers	Separate SIS elements in R-103 (two per cell, 2oo3 per hot zone) (TSHH-107S, 220 °C)	SIS	yes	PK-10	
SL-R6	Safety layers	V-104 level low-low closes a new SIS valve XV-104 on the product line (LSLL-104, 5 % (assumed))	SIS	yes	PK-10	
SL-R7	Safety layers	Stop P-100A/B on unit trip (I-P100, on every unit trip)	SIS	yes	PK-10	
SL-R8	Safety layers	Safety valve on the R-101 overhead (no isolation between it and R-101) (PSV-101, 36 bar g (assumed design pressure))	HW	yes	PK-15	
SL-R9	Safety layers	Safety valve on the R-102 off-gas line (PSV-102, 36 bar g (assumed))	HW	yes	PK-15	
SL-R10	Safety layers	Safety valve on V-104 (PSV-104, 36 bar g (assumed))	HW	yes	PK-15	
SL-R11	Safety layers	Safety valve on K-101 discharge (upstream of FV-103) (PSV-112, 42 bar g (assumed))	HW	yes	PK-15	
SL-R12	Safety layers	Safety valve on steam drum D-101 (PSV-109, 8 bar g (assumed))	HW	yes	PK-15	
SL-R13	Safety layers	Thermal relief on the E-103 cooling-water side (TSV-113, 10 bar g (assumed))	HW	yes	PK-15	
SL-R14	Safety layers	Thermal relief on the E-106 cooling-water side (TSV-106, 10 bar g (assumed))	HW	yes	PK-15	
SL-R15	Safety layers	Thermal relief on the R-102 jacket (TSV-102J, 10 bar g (assumed))	HW	yes	PK-15	
SL-R16	Safety layers	Thermal relief on each catalyst filter F-102A/B (TSV-102F, 36 bar g (assumed))	HW	yes	PK-15	
SL-R17	Safety layers	Emergency isolation valve on benzene at battery limit (EIV-100, fail closed, fire-safe)	HW	yes	PK-15	
SL-R18	Safety layers	EIV on the R-101 bottom outlet (pump-around suction) (EIV-101, fail closed, fire-safe)	HW	yes	PK-15	
SL-R19	Safety layers	EIV on the R-102 bottom outlet (EIV-102, fail closed, fire-safe)	HW	yes	PK-15	
SL-R20	Safety layers	EIV on the V-104 product line (EIV-104, fail closed, fire-safe)	HW	yes	PK-15	
SL-R21	Safety layers	Check valve at P-100A/B discharge (NRV-100)	HW	yes	PK-15	
SL-R22	Safety layers	Check valve at K-101 discharge (NRV-112)	HW	yes	PK-07	
SL-R23	Safety layers	Flare knock-out drum (V-190)	HW	yes	PK-14	
SL-R24	Safety layers	Kerbed area round the benzene pumps to the oily-water sewer (BUND-1)	HW	yes	PK-14	
SL-R25	Safety layers	Kerbed slab under the reactor structure to a closed drain (BUND-2)	HW	yes	PK-14	
SL-R26	Safety layers	Raney-Ni handled under liquid and nitrogen (catalyst pot) (CAT-1, N2 blanket)	DOC	yes	PK-13	

ID	Source	Recommendation	Disposition	Before start-up	Package	Closed / sign
SL-R27	Safety layers	Spent catalyst from F-102 kept wet in closed drums (CAT-2)	DOC	yes	PK-13	
SL-R28	Safety layers	Fireproofing of the R-103 skirt and V-104 saddles (FP-1, 2 h (assumed))	HW	yes	PK-14	
SL-R29	Safety layers	Fire-water ring main (FW-1, 10 bar g (assumed))	HW	yes	PK-17	
SL-R30	Safety layers	Hydrant west of the reactors (HYD-1)	HW	yes	PK-17	
SL-R31	Safety layers	Hydrant east of V-104 (HYD-2)	HW	yes	PK-17	
SL-R32	Safety layers	Fire-water monitor on the reactor structure (MON-1)	HW	yes	PK-17	
SL-R33	Safety layers	Fire-water monitor at V-104 / K-101 (MON-2)	HW	yes	PK-17	
SL-R34	Safety layers	Deluge on K-101 (DEL-101, opens on confirmed fire (assumed))	HW	yes	PK-17	
SL-R35	Safety layers	Water spray on V-104 (DEL-104, opens on confirmed fire (assumed))	HW	yes	PK-17	
SL-R36	Safety layers	Hydrogen detector at K-101 (GD-1, 10 % / 25 % LFL (assumed))	HW	yes	PK-17	
SL-R37	Safety layers	Hydrogen detector at the R-101 top flanges (GD-2, 10 % / 25 % LFL (assumed))	HW	yes	PK-17	
SL-R38	Safety layers	Hydrocarbon (benzene) detector at P-100A/B (GD-3, 10 % / 25 % LFL (assumed))	HW	yes	PK-17	
SL-R39	Safety layers	UV/IR flame detector covering K-101 and V-104 (FD-1)	HW	yes	PK-17	
SL-R40	Safety layers	Manual call point (MCP-1)	HW	yes	PK-17	
SL-R41	Safety layers	Site siren / PA (SIREN)	HW	yes	PK-17	
SL-R42	Safety layers	Windsock (WIND)	HW	yes	PK-17	
SL-R43	Safety layers	Muster point (MUSTER)	HW	yes	PK-17	
SL-R44	Safety layers	Safety shower and eye wash at the benzene pumps (SHOWER)	HW	yes	PK-17	

6 · Commissioning and start-up steps ↔ SOPs and OTS procedures

Each step names its SOP (SOP register of the OTS) and the steps of the graded procedure that check it on the simulator. Times are h:mm from loading the procedure in the Transition tab's reference runs: cold startup "+4.35 t/h every 5 min (≈ 0.9 t/h per min)" (first feed 0:53, switch to COOL 1:02, design rate 1:37, on spec 1:56, limit 3:30; 5.4 t to slop) and the normal shutdown (all steps at 0:32, limit 2:00).

Step	What is done	Subsystem	SOP	Graded procedure step(s) on the OTS	Reference run	Sign
Commissioning (no hydrocarbon)						
C-01	Utilities commissioned and stable (CW, MP steam, BFW, nitrogen, instrument air, power, flare)	S10.1	SOP-CYC-01	— (not a graded step)	—	
C-02	Loop inerted: nitrogen pressure-purge cycles until AI-100 is below the hydrogen permissive	S30.4	SOP-CYC-01	startup 1 · Loop pressure-purged with nitrogen (HV-120 / HV-121): AI-100 below 1 % O2	0:02 (+2.3 min)	
C-03	Final nitrogen tightness test of the reinstated loop (LT-01; not simulated)	S30.1	SOP-CYC-01	— (not a graded step)	—	
Start-up (first hydrocarbon, RFSU)						
C-04	Start-up cyclohexane charged to R-101 through HV-116	S40.1	SOP-CYC-02	startup 2 · R-101 charged with start-up cyclohexane to 50 % or more and HV-116 closed	0:25 (+23.5 min)	
C-05	Agitator M-101 started; Raney-Ni charged from the pot through HV-117	S40.1	SOP-CYC-07, SOP-CYC-02	startup 3 · Agitator M-101 running and the Raney-Ni catalyst charged (HV-117 closed again)	0:30 (+5.0 min)	
C-06	Cooling water to E-103; D-101 filled on LIC-109, PIC-110 in AUTO	S10.3	SOP-CYC-10	startup 4 · Utilities in: HV-113 CW open, LIC-109 in AUTO with D-101 level 30 % or more, PIC-110 in AUTO	0:30 (+0.0 min)	
C-07	Hydrogen admitted (XV-105) and the loop pressured on PIC-105	S30.1	SOP-CYC-02	startup 5 · Hydrogen admitted (XV-105 open) with PIC-105 in AUTO at 29 bar g	0:30 (+0.0 min)	
C-08	K-101 started; hydrogen circulating through the R-101 sparger	S30.2	SOP-CYC-02	startup 6 · K-101 running (suction above 10 bar g) and FIC-103 in AUTO at 15 000 Nm³/h or more	0:31 (+1.0 min)	
C-09	R-101 heated on MP steam through E-101 to the light-off temperature	S40.2	SOP-CYC-10	startup 7 · R-101 heated on MP steam (HS-101 HEAT, HV-110 open, pump-around running) to 140 °C or more with the loop at 28–30 bar g	0:50 (+19.1 min)	
C-10	First benzene: XV-101 permitted, P-100A, FIC-101 at a low rate — light-off	S20.1	SOP-CYC-02	startup 8 · Benzene permitted (XV-101) and FIC-101 in AUTO at 6–12 t/h	0:54 (+3.2 min)	
C-11	R-102, filters, R-103 and V-104 in line; level and R-102 temperature loops in AUTO	S50.2	SOP-CYC-02	startup 9 · Levels in AUTO: LIC-101, LIC-102 (P-102 running), LIC-104; TIC-102 in AUTO	1:02 (+8.5 min)	
C-12	E-101 switched from HEAT to COOL; TIC-101 → FIC-104 cascade; LP steam export starts	S40.2	SOP-CYC-10	startup 10 · E-101 on COOL with HV-110 closed; TIC-101 AUTO at 200 °C and FIC-104 in CASCADE	1:02 (+0.0 min)	
C-13	Benzene and hydrogen raised together to the design rate	S20.1	SOP-CYC-02, SOP-CYC-05	startup 11 · Design rate: FIC-101 at 34 t/h or more and FIC-103 at 44 000 Nm³/h or more	1:38 (+35.5 min)	
C-14	Product on specification; HS-115 to TK-106	S60.4	SOP-CYC-02	startup 12 · Product on spec (AI-103 ≤ 10, AI-104 ≤ 200 ppm) and HS-115 switched to TK-106	1:56 (+17.9 min)	
C-15	Line-out held: start-up complete, unit handed to normal operation for the test run	S40.1	SOP-CYC-02, SOP-CYC-13	startup 13 · Hold the line-out 10 min: TI-100 195–205 °C, product on spec, PIC-105 28–30 bar g	2:06 (+10.0 min)	
Initial operation (before the performance test)						
Step	**What is done**	**Subsystem**	**SOP**	**Graded procedure step(s) on the OTS**	**Reference run**	**Sign**
C-16	Spare equipment proven on line: pump change-over, including a running-pump trip	S40.2	SOP-CYC-09	esd 1 · Spare pump P-101B running esd 2 · FIC-104 in CASCADE with the flow back above 250 t/h	—	

Step	What is done	Subsystem	SOP	Graded procedure step(s) on the OTS	Reference run	Sign
				esd 3 · R-101 back within 195–205 °C (TI-100) esd 4 · Hold 5 min: TI-100 195–205 °C, product on spec, no trip esd 5 · P-101A reported to maintenance		
C-17	Catalyst filter change-over F-102A → F-102B proven	S50.2	SOP-CYC-08	filt 1 · F-102 switched to the clean filter B filt 2 · PDI-102 back below 0.5 bar with LIC-102 in AUTO filt 3 · R-102 level back within 45–65 % filt 4 · F-102A reported for cleaning	—	
C-18	Rate change demonstrated above design (SOP-CYC-05 limits)	S20.1	SOP-CYC-05	Rate increase with an unknown upset (graded; the steps appear on the DCS console when the procedure is loaded)	—	
C-19	Trip response and restart after a trip (if a trip occurs, or as a planned SIS demonstration)	S70.2	SOP-CYC-04, SOP-CYC-11	— (not a graded step)	—	
Performance test and handover						
C-20	Performance test at design (part 10)	S60.4	SOP-CYC-13, SOP-CYC-06	— (not a graded step)	—	
C-21	First planned shutdown (catalyst inspection), after the test	S40.1	SOP-CYC-03, SOP-CYC-12	shutdown 1 · Rate cut to half: FIC-101 at 18 t/h or less and FIC-103 at 25 000 Nm³/h or less shutdown 2 · Benzene stopped: FIC-101 in MAN at 0 %, P-100A/B stopped, XV-101 closed shutdown 3 · Reaction finished with hydrogen still circulating: make-up FI-106 below 3 000 Nm³/h shutdown 4 · Product to slop (HS-115 → TK-107) and R-102 jacket on COOL (HS-102) shutdown 5 · R-101 cooled below 130 °C (lower PIC-110 and the TIC-101 set point) shutdown 6 · K-101 stopped, XV-105 closed, PIC-105 in MAN at 0 % shutdown 7 · Loop depressured to flare to 1 bar g or less, then the vent closed shutdown 8 · Nitrogen purge: recycle gas below 4 % hydrogen (AI-105) shutdown 9 · Agitator and pumps stopped (M-101, P-101A/B, P-102A/B)	0:32 (all steps)	

7 · Performance test

Design rate 34.8 t/h benzene. Window: 72 h at design in the plant (typical); 10–60 min of simulated time on the OTS.

ID	Criterion	Tag	Design (model)	Acceptance	Basis	Result
PT-01	Benzene feed rate (capacity)	FI-101	34.80 t/h	≥ 34.63 t/h	design rate; 0.5 % allowance for meter uncertainty (typical)	
PT-02	Cyclohexane product to TK-106	FI-110	37.50 t/h	≥ 37.31 t/h	design product (300 kt/y at 8000 h/y); 0.5 % allowance (typical)	
PT-03	Benzene in product	AI-103	2.0 ppm	≤ 10.0 ppm	product specification (alarm limit AAH-103)	
PT-04	Methylcyclopentane (MCP) in product	AI-104	60 ppm	≤ 200 ppm	product specification (alarm limit AAH-104)	
PT-05	Hydrogen utilisation of the make-up	FI-106, FI-107, AI-105	94.16 %	≥ 93.16 %	1 − purge H_2 / make-up H_2 (make-up 99.5 % H_2); design − 1 percentage point (assumed)	
PT-06	Specific make-up hydrogen	FI-106 / FI-110	852.7 Nm^3/t	≤ 869.7 Nm^3/t	design + 2 % (assumed)	
PT-07	LP steam export	FI-111	14.72 t/h	≥ 14.28 t/h	design − 3 % (assumed); D-101 on PIC-110 at its design set point	
PT-08	Electrical power, absorbed	model (pumps, agitator, K-101)	1,633 kW	≤ 1,682 kW	design + 3 % (assumed); measured at the MCC on the plant	
PT-09	R-101 temperature during the test	TI-100	200.0 °C	195.0 – 205.0 °C	line-out band of the graded cold startup (PROCS)	
PT-10	Loop pressure during the test	PIC-105	29.00 bar g	28.00 – 30.00 bar g	line-out band of the graded cold startup (PROCS)	
PT-11	Recycle hydrogen purity	AI-105	92.1 %	≥ 85.0 %	low alarm AAL-105	
PT-12	No SIS trip during the test window	SIS	none	none	unit stays in operation	

- Plant lined out at design for at least 24 h before the test (typical); no set-point changes during the window.
- Average each measurement over the window; laboratory samples of feed, make-up gas and product at the start, middle and end (typical).
- A criterion passes when its window average meets the limit; the operating conditions (PT-09 … PT-11) must hold for the whole window and no trip may occur.

Worked example. Hydrogen utilisation = 1 − purge H_2 / make-up H_2 = 1 − FI-107·AI-105 / (FI-106·(1 − y_{inert})); at design 94.16 % (acceptance ≥ 93.16 %). The tab of the OTS computes every criterion on the running simulator (instant reading or a window average) and on a copy of the design state.

8 · Handover

Care, custody and control pass to operations system by system with these records and an agreed punch-B list; provisional acceptance follows a passed performance test, final acceptance the warranty period (typical practice).

Record	Content	Handed over / sign
Mechanical completion certificates	One per subsystem of part 5 (S10.1 … S70.3)	
Pre-commissioning records	Flushing, test packs TP-01 … TP-06, leak tests, drying, loop checks, motor runs, catalyst loading	
SIS function-test records	SFT and ALT sheets of part 7 signed; proof-test intervals of the SRS entered in the maintenance system	
PSSR record	Part 8 with the recommendation close-out	
As-built documents	P&ID CHX-V1-PID-001, C&E CHX-V1-IN-CE-001, SRS CHX-V1-IN-SRS-001, instrument list, line list, equipment datasheets	
Vendor data books	Per package of CHX-V1-PC-PKG-001: certificates, FAT reports, manuals, spare-parts lists	
Operating procedures and training records	SOP register (13 SOPs); simulator grades of every console operator	
Performance test report	Part 10 results with the averaging window and the laboratory analyses	
Punch-B list and open recommendations	Items that may follow, each with an owner and a date; the "after start-up" recommendations with their interim measures	
Catalyst records	R-103 loading record; Raney-Ni lot certificates and charge record	

9 · Engineering queries

EQ	Query	Documents	Proposed resolution
EQ-ec-1	R-101 and R-102 are both slurry reactors, but the model has an agitator motor on R-101 only (MOTORS: M-101) and the P&ID shows no mixer on R-102. How R-102 keeps the Raney-Ni suspended is not defined.	PK-01; equipment list CHX-V1-ME-EQL-001; load list CHX-V1-EE-LL-001	Process to state the R-102 mixing (agitator or gas-induced). If agitated, add M-102 to PK-01, the load list and the motor solo runs.
EQ-ec-2	In HEAT mode (HS-101) MP steam at 10 bar g reaches the E-101 shell, which is the D-101 steam side protected by PSV-109 at an assumed 8.0 bar g (Safety layers SL-R12). The shell, the drum and TP-05 have no agreed design pressure.	PK-03, PK-04, TP-05; Safety layers SL-R12; relief load basis CHX-V1-PE-RLB-001	Either design the E-101 shell and D-101 for the MP steam pressure (PSV-109 set to suit) or add pressure reduction and relief on the MP steam to E-101. Then fix TP-05.
EQ-ec-3	The Raney-Ni catalyst pot on R-101 (HV-117) has no equipment number on the P&ID and no datasheet, yet SOP-CYC-07 pressures it with nitrogen above the R-101 pressure.	P&ID CHX-V1-PID-001; PK-05; SOP-CYC-07	Give the pot an equipment number and a datasheet; design pressure at least the loop design pressure; add it to test pack TP-01.
EQ-ec-4	The SIS scope is not frozen. Open recommendations add sensors, logic and valves: HAZOP R01, R03–R07, R14, R16, R19, R24 and R26; Safety layers SL-R3 … SL-R7; the second make-up valve of the LOPA and SRS.	PK-10; C&E CHX-V1-IN-CE-001; SRS CHX-V1-IN-SRS-001	Close these in the HAZOP close-out before the SIS PO (week 18 in part 4). Order with spare I/O (about 20 %, typical). Re-issue the C&E, then the SIS FAT procedure.
EQ-ec-5	Raney-Ni usually comes water-wet. SOP-CYC-07 rinses with cyclohexane but has no step to replace the water. Water charged into R-101 at 200 °C flashes, then ends up in V-104 and the product.	PK-13; SOP-CYC-07; checklist CL-03	Specify the carrier liquid in PK-13: either supplied under cyclohexane, or a solvent exchange in the pot as a written step in SOP-CYC-07.
EQ-ec-6	The relief valves (SL-R8 … SL-R16) are recommended additions with assumed set pressures and no orifice sizes. PK-15 cannot be bought, and MC-04 and PSSR item PS-03 cannot be closed, until the relief load basis is issued.	PK-15; MC-04; PS-03; Relief & vent tab CHX-V1-PE-RLB-001	Issue the relief load basis before week 24 (the PK-15 PO week); set pressures follow the design pressures in EQ-ec-2.
EQ-ec-7	The design pressure of the benzene feed line (TP-04) is assumed equal to the loop (36 bar g). The P-100A/B shut-off head with a closed outlet may be higher.	PK-06; TP-04; line list CHX-V1-PE-LL-001	Take the P-100 shut-off pressure from the pump datasheet. Set the line design pressure, and the TP-04 test pressure, from it.